

Economics 5040: Dynamic Stochastic Modeling

Problem Set 2: Expected Utility and Asset Pricing

Instructions

Submit one completed Jupyter notebook containing your derivations, Python code, figures, numerical answers, and economic explanations. Derive results analytically when possible and then use Python to calculate or verify them. You may consult AI tools, but you are responsible for understanding every result you submit.

Problem 1. CARA Utility and the Price of Risk

Let wealth be w and suppose von Neumann–Morgenstern utility is

$$u(w) = \frac{1 - e^{-\gamma w}}{\gamma}.$$

This is a constant absolute risk aversion (CARA) utility function.

- (a) Show analytically that the coefficient of absolute risk aversion is

$$A(w) = -\frac{u''(w)}{u'(w)} = \gamma,$$

and that the coefficient of relative risk aversion is

$$\rho(w) = -\frac{wu''(w)}{u'(w)} = \gamma w.$$

Explain why the name CARA is appropriate.

- (b) Set $\gamma = 1/1000$. Plot $u(w)$ for

$$0 \leq w \leq 100,000.$$

Then make a second, economically useful comparison of curvature for several positive values of γ . Choose plotting ranges that actually reveal the differences and explain your choices.

- (c) Initial wealth is $w_0 = 30,000$ and $\gamma = 1/1000$. Consider a 50–50 gamble that loses 100 or wins 110. Compute expected utility from accepting the gamble and utility from refusing it. Show numerically which choice is preferred. Compute the certainty equivalent of final wealth and the risk premium.

- (d) The Devil offers you all the wealth in the universe, now and forever, in exchange for accepting a 50–50 gamble that may cost you 1000. Interpret “all the wealth in the universe” using

$$\lim_{w \rightarrow \infty} u(w).$$

Compute the limit and determine whether the offer is accepted when $\gamma = 1/1000$. Then repeat when the possible loss is only 500. Be precise.

- (e) Explain what the Devil’s Offer teaches us about CARA preferences. Why can bounded utility make an arbitrarily large upside insufficient compensation for a sufficiently bad downside?

Problem 2. From Outcomes to Expected Utility

There are three consumption outcomes

$$x_1 = 25, \quad x_2 = 100, \quad x_3 = 225.$$

A consumer has von Neumann–Morgenstern utility

$$u(c) = \log c.$$

- (a) Normalize utility so that

$$u(x_1) = 0, \quad u(x_3) = 1.$$

For x_2 , find the probability q such that receiving x_2 for certain is indifferent to receiving x_3 with probability q and x_1 otherwise.

- (b) Let L put probabilities

$$(0.25, 0.50, 0.25)$$

on (x_1, x_2, x_3) . Reduce L to an equivalent lottery involving only x_1 and x_3 . Verify that the probability on x_3 equals expected normalized utility.

- (c) Construct another lottery M of your choosing. Rank L and M using expected utility, and verify that a positive affine transformation preserves the ranking.
- (d) Apply a strictly increasing but non-affine transformation to the outcome utility index. Find, by computation if necessary, an example showing that expected-utility rankings need not be preserved. Explain why this differs from ordinary ordinal utility.

Problem 3. State Prices and Arrow Securities

There are two dates. At date 1 there are two equally likely states, H and L . Preferences are logarithmic with $\beta = 0.96$. Suppose

$$c_0 = 10, \quad c_1(H) = 15, \quad c_1(L) = 5.$$

- (a) Derive and compute the state prices q_H and q_L from the consumer’s first-order conditions.
- (b) Explain why $q_L > q_H$ even though the states are equally likely. Compute

$$\frac{q_H}{\pi_H}, \quad \frac{q_L}{\pi_L},$$

and interpret these probability-adjusted state prices.

- (c) Hold c_0 and $c_1(H)$ fixed and vary $c_1(L)$ from 2 to 15. Plot

$$\frac{q_L}{q_H}.$$

Explain the graph economically.

- (d) Compute the stochastic discount factor m in each state and verify

$$q_s = \pi_s m_s.$$

Problem 4. Assets as Bundles of State-Contingent Claims

Continue with Problem 3. Consider the assets

$$A = (1, 1), \quad B = (2, 0), \quad C = (0, 2), \quad D = (3, 1),$$

where each vector lists date-1 payoffs in states (H, L) .

- (a) Price each asset using the state prices.
- (b) Calculate each asset's state-contingent gross returns and expected gross return.
- (c) Asset A is risk free. Compute R_f and verify

$$\frac{1}{R_f} = q_H + q_L.$$

- (d) Compare B and C . Which has the higher expected return? Explain why using the states in which the assets pay.
- (e) Verify for all four assets that

$$p = E[mx]$$

and

$$1 = E[mR].$$

Problem 5. What Kind of Risk Commands a Premium?

Use the same economy.

- (a) For assets B , C , and D , compute

$$\text{Cov}(m, R).$$

- (b) Verify numerically that

$$E[R] - R_f = -R_f \text{Cov}(m, R).$$

- (c) Construct an asset E with the same expected payoff as D but with more of its payoff in the low-consumption state. Compare its price and expected return with D .
- (d) Construct another asset F with the same expected payoff as D but with more payoff in the high-consumption state. Compare D , E , and F .
- (e) Explain in a short paragraph why volatility alone is not the relevant measure of priced risk in this economy.

Problem 6. Complete and Incomplete Markets

There are three possible states tomorrow.

- (a) With one Arrow security for each state, write the 3×3 payoff matrix and use Python to verify that it has rank 3. Show how an arbitrary payoff vector can be replicated.

- (b) Now suppose only a risk-free bond

$$b = (1, 1, 1)$$

and an asset

$$a = (2, 1, 0)$$

are traded. Form the payoff matrix and compute its rank.

- (c) Can

$$x = (0, 1, 0)$$

be replicated? Solve exactly if possible; otherwise use least squares and report the residual.

- (d) Characterize algebraically the set of payoff vectors attainable with these two assets.

- (e) Explain why this market is incomplete and identify the state-specific insurance that is missing.

Problem 7. Synthesis

- (a) In approximately 400–500 words, explain how the idea of a relative price has developed from Week 1 to Week 2. Connect dated commodities, state-contingent commodities, Arrow securities, state prices, the stochastic discount factor, the risk-free rate, and complete markets.
- (b) End your notebook with a short section entitled

What surprised me?

Identify one result from this assignment that was not obvious to you before doing the computations. Explain why it surprised you and what you now understand differently.